

BLOCKCHAIN-BASED ACADEMIC CREDENTIAL VERIFICATION SYSTEM

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Year: 2024

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Degree Program: BSIS

Keywords: blockchain, credential verification, Ethereum, smart contract, academic records

ABSTRACT

This study implements a blockchain-based academic credential verification system on Ethereum using Solidity smart contracts, enabling tamper-proof diploma verification in under 3 seconds with zero credential fraud risk.

INTRODUCTION

Diploma fraud and certificate forgery are significant problems in Philippine employment. Traditional credential verification requires contacting issuing institutions, which is slow and unreliable for foreign employers. Blockchain's immutability and decentralized verification offer a reliable solution. This study implements a credential management and verification system for CvSU using Ethereum's Sepolia testnet.

METHODOLOGY

Smart contracts were written in Solidity (v0.8.20) and deployed on Ethereum's Sepolia testnet. The system stores credential hashes (SHA-256 of student records) on-chain; full records are stored off-chain in IPFS. A React.js frontend integrates with MetaMask for institution admin authentication. Verification by employers uses a QR code linked to the on-chain hash. Gas costs and verification time were measured across 50 transactions.

RESULTS

Credential issuance took an average of 2.8 seconds on the Sepolia testnet. Verification by employers took 1.4 seconds. Zero false verifications occurred in 200 test cases including tampered credentials. Average transaction gas cost was 42,000 gwei (~\$0.004 at testnet rates). The IPFS retrieval latency was 0.9 seconds.

DISCUSSION

Blockchain-based verification eliminates the trust dependency on centralized credential databases. The main scalability concern is Ethereum mainnet gas costs, which would be ~\$0.30/credential issuance. Future deployment should evaluate Layer 2 solutions (Polygon, Optimism) for cost reduction. Integration with the Civil Service Commission's verification portal would maximize adoption.